

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A

Comparative Analysis

Code of Conduct:

I R Dinesh-192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R Dinesh

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant	
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison	

Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided	
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation	
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A 2.0 2.5

B 3.0 3.8

C 4.0 5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

2. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

3. Precision-Recall Trade-off Analysis

Two models produce:

Model TP FP FN

A 70 30 20

B 85 25 25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

4. F1-Score Comparison

Two models:

Model Precision Recall

A 0.8 0.6

B 0.75 0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

5. Optical Flow Consistency

Motion values:

Frame Method A Method B

1 4 6

2 5 8

3 6 9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

1. Depth Estimation Comparison – Stereo vs SfM

Given:

Object	Stereo Vision	SfM	Actual Depth
A	2.0	2.5	2
B	3.0	3.8	3
C	4.0	5.0	4

(a) Absolute Error

Formula:

Object	Stereo Error	SfM Error
A	(	2.0-2
B	(	3.0-3
C	(	4.0-4

Total Error:

- Stereo Vision = **0 m**
- SfM = **2.3 m**

Average Error:

- Stereo Vision = **0 m**
- SfM = **0.77 m**

(b) Comparison

Stereo Vision is more accurate because its average absolute error is **0 m**, while SfM has an average error of approximately **0.77 m**.

Final Answer:

Stereo Vision is the better method for accurate depth estimation.

2. Pattern Recognition Accuracy Comparison

Given:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) Accuracy

Formula:

Traditional:

Deep Learning:

(b) Comparison

System	Accuracy
Traditional	85%
Deep Learning	92%

Final Answer:

Deep Learning performs better because its accuracy is **92%**, which is higher than the Traditional system's **85%**.

3. Precision-Recall Trade-off Analysis

Given:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Precision and Recall

Formulas:

Model A

Precision:

Recall:

So:

- Precision = **70%**
- Recall = **77.8%**

Model B

Precision:

Recall:

So:

- Precision = **77.3%**
- Recall = **77.3%**

(b) Comparison

Model	Precision	Recall
A	70%	77.8%
B	77.3%	77.3%

Final Answer:

Model B is more balanced because its precision and recall are almost equal and both are higher than Model A's precision.

4. F1-Score Comparison

Given:

Model	Precision	Recall
A	0.8	0.6
B	0.75	0.75

Formula:

Model A

Model B

Comparison

Model	F1-Score
A	0.686

Model	F1-Score
B	0.75

Final Answer:

Model B is better because it has the higher F1-score of **0.75**, indicating a better balance between precision and recall.

5. Optical Flow Consistency

Given:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Average Motion

Method A:

Method B:

(b) Compare Consistency

Method A:

- Values = 4, 5, 6
- Range =

Method B:

- Values = 6, 8, 9
- Range =

A smaller range indicates more consistent motion.

Method	Average Motion	Range
A	5.00	2
B	7.67	3

Final Answer:

Method A is more stable and consistent because it has a smaller variation/range of **2**, compared with Method B's **3**.